



Customer Shopping Behavior Analysis

Uncovering insights into spending patterns, customer segments & product preferences

3,900 TRANSACTIONS

PYTHON

MYSQL

POWER BI

What This Project Is About

Analyzed 3,900 retail transactions across multiple product categories through a full pipeline: Data Cleaning → SQL Analysis → Interactive Dashboard. Goal: uncover actionable insights to guide strategic business decisions.

Spending Patterns

Customer
Segmentation

Product & Subscription Behavior



Data & Tech Stack



Dataset

- 3,900 rows · 18 columns
- 37 missing values in Review Rating (fixed via median imputation)
- Features: Demographics, Purchase Details, Shopping Behavior

Tools Used

Python (pandas)

Cleaning & feature engineering

MySQL Workbench

Business SQL queries

Power BI

Interactive dashboard

Jupyter Notebook

EDA workflow

Cleaning & Preparing the Data

01

Load & Explore

Loaded dataset with pandas; explored using `df.info()` and `.describe()`

03

Standardize Columns

Renamed all columns to `snake_case` for consistency

05

Drop Redundancy

Dropped `promo_code_used` (redundant with `discount_applied`)

02

Impute Missing Values

Imputed 37 missing Review Ratings using median per product category

04

Engineer Features

Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase records

06

Load to MySQL

Loaded cleaned data into MySQL for SQL analysis

Where Is the Money Coming From?

Gender Revenue Split

Male customers: \$157,890 revenue
vs. Female: \$75,191

Shipping Spend

Express shipping avg spend:
\$60.48 vs. Standard: \$58.46

Discount Users

839 discount users still spent
above the average purchase
amount

Subscriber Spend

Subscribers avg spend:
\$59.49 · Non-subscribers: \$59.87

Total Revenue by Status

Non-subscribers generate far more
total revenue: \$170,436 vs.
\$62,645



Key Takeaway: Spend behavior is nearly equal across groups — volume is the differentiator.

What Are Customers Buying?

Top 5 Rated Products

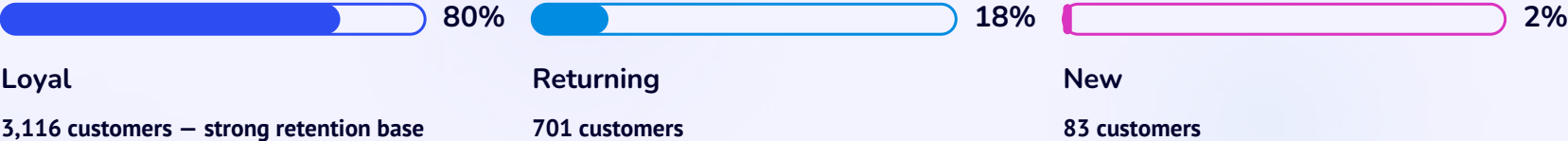


Best-Selling per Category



Who Are Our Customers?

Customer Segments



Most Discount-Dependent Products

Product	Discount Rate
Hat	50%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

 Repeat Buyers: Among customers with 5+ purchases – 2,518 are non-subscribers vs. only 958 subscribers

Which Age Group Drives the Most Revenue?



Revenue by Age Group

Young Adults lead with \$62,143 in total revenue, followed closely by Middle-aged customers at \$59,197. The gap between age groups is relatively small, but Young Adults consistently outperform all others.



Key Takeaway: Young Adults are the #1 revenue contributors – the highest priority segment for targeted campaigns.

Interactive Dashboard Overview

KPI Cards

3.9K

Total Customers

\$59.76

Avg Purchase Amount

3.75

Avg Review Rating

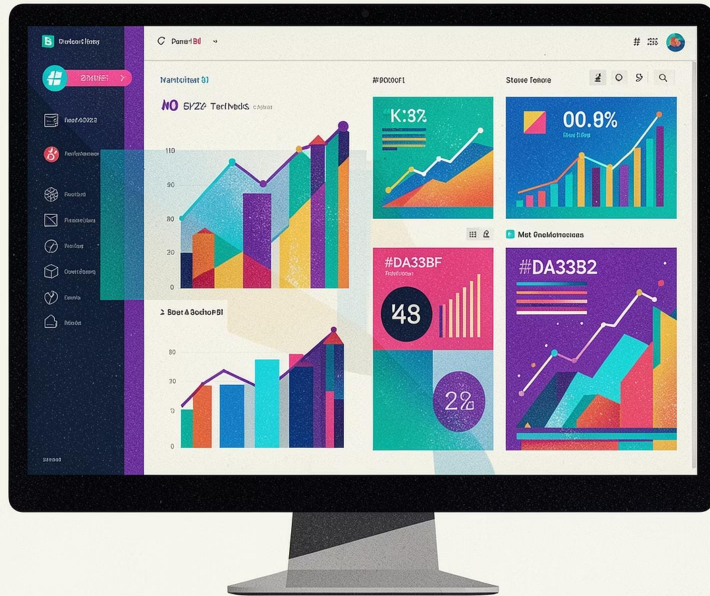
Visuals

→ Donut Chart – Subscription status: Yes 27% / No 73%

→ Bar Charts – Revenue & Sales by Category

→ Horizontal Bars – Revenue & Sales by Age Group

→ Slicers – Gender, Category, Subscription Status, Shipping Type



What Should the Business Do Next?

1

Boost Subscriptions

73% are non-subscribers; promote exclusive perks to convert them

2

Launch Loyalty Programs

Move Returning customers into the Loyal segment

3

Review Discount Policy

Hat, Sneakers & Coat discounted ~50% of the time; protect margins

4

Spotlight Top Products

Feature Gloves, Sandals & Boots (highest rated) in campaigns

5

Target Young Adults

#1 revenue group; prioritize in marketing spend

6

Upsell Express Shipping

Express users spend more; promote at checkout



Key Takeaway: A data-driven approach across segmentation, discounts, and product strategy can directly improve revenue, retention, and customer satisfaction.